



SOLUTIONS OF FUNDAMENTALS OF MATHEMATICS-I

EXERCISE # 1

PART-1

Section (A)

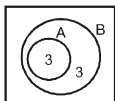
- A-1.** Collection of all intelligent women in Jalandhar is not a set as it is not a well defined collection. It is not possible to decide logically which woman is to be included in the collection and which is not to be included.
- A-2.** 2, 3, 5 and 7 are the only positive primes less than 10.
- A-3.** Obvious
- A-4.** (i) $x^2 - 1 = 0$ $x = \pm 1$
 (ii) $x^2 + 1 = 0$ $x = \pm i$ $x \in \phi$
 (iii) $x^2 - 9 = 0$ $x = \pm 3$
 (iv) $x^2 - x - 2 = 0$, $x = 2, -1$

- A-5.** $P(A) = \{\phi, \{\phi\}, \{\{\phi\}\}, \{\phi, \{\phi\}\}\} = \{\phi, \{\phi\}, \{\{\phi\}\}, A\}$

Section (B)

- B-1.** $A = \{1, 2, 3\}$
 $B = \{3, 4\}$
 $C = \{4, 5, 6\}$
 $B \cap C = \{4\}$
 $B \cup C = \{3, 4, 5, 6\}$
 $A \cup (B \cap C) = \{1, 2, 3, 4\}$
 $A - (B \cap C) = \{1, 2, 3\}$
 $(B \cup C) - A = \{4, 5, 6\}$
- B-2.** Obvious
- B-3.** $bN \cap cN$
 (+ve integral multiple of b) $\cap$ (+ve integral multiple of c)
 since b & c are relatively primes $= b \cdot c \cdot N$
 $\therefore d = bc$

B-4.



for minimum $n(A \cup B)$, the set A is subset of B and for maximum $n(A \cup B)$, the sets A and B are disjoint set.

$$\begin{aligned} \text{also } n(A \cup B) &= n(A) + n(B) - n(A \cap B) \\ \Rightarrow \text{minimum } n(A \cap B) &= 0, \text{ maximum } n(A \cap B) = 3 \\ \Rightarrow \text{minimum value of } n(A \cup B) &= 3 + 6 - 3 = 6 \text{ or maximum value of } n(A \cup B) = 3 + 6 - 0 = 9 \end{aligned}$$

Section (C)

- C-1.** $n(A^c \cap B^c) = n[(A \cup B)^c] = n(U) - n(A \cup B) = n(U) - [n(A) + n(B) - n(A \cap B)]$
 $= 700 - [200 + 300 - 100] = 300.$
- C-2.** Let number of newspapers is x . As every newspaper is read by 60 students
 Since, every students reads 5 newspapers
 $\therefore 60x = 300(5) \Rightarrow x = 25.$





- C-3.** $n(A) = 40\% \text{ of } 10,000 = 4,000$
 $n(B) = 20\% \text{ of } 10,000 = 2,000$
 $n(C) = 10\% \text{ of } 10,000 = 1,000$
 $n(A \cap B) = 5\% \text{ of } 10,000 = 500$
 $n(B \cap C) = 3\% \text{ of } 10,000 = 300$
 $n(C \cap A) = 4\% \text{ of } 10,000 = 400$
 $n(A \cap B \cap C) = 2\% \text{ of } 10,000 = 200$
 $n(A \cap B^c \cap C^c) = n[A \cap (B \cup C)^c]$
 $= n(A) - n[A \cap (B \cup C)] = n(A) - n[(A \cap B) \cup (A \cap C)]$
 $= n(A) - [n(A \cap B) + n(A \cap C) - n(A \cap B \cap C)]$
 $= 4000 - [500 + 400 - 200] = 4000 - 700 = 3300.$
- C-4.** $n(A) = 21, n(B) = 26, n(C) = 29$
 $n(A \cap B) = 14, n(A \cap C) = 12, n(B \cap C) = 13, n(A \cap B \cap C) = 8$
 $n(C \cap A' \cap B') = n(C \cap A' \cap B') = n(C) - n[(C \cap A) \cup (C \cap B)]$
 $n(C) - [n(C \cap A) + n(C \cap B) - n(A \cap B \cap C)]$
 $29 - [12 + 13 - 8] = 12$
 $n(A \cap B \cap C') = n(A \cap B) - n(A \cap B \cap C) = 14 - 8 = 6$

Section (D)

- D-1.** (i) $\frac{+}{-3} \frac{-}{-2} \frac{+}{1} \frac{-}{3} \frac{+}{3} \Rightarrow x \in (-3, -2) \cup (1, 3)$
- (ii) $\frac{-}{-9} \frac{ND}{-2} \frac{0}{1} \frac{0}{8} \frac{+}{8} \frac{ND}{-} \Rightarrow x \in \{-2\} \cup [1, 8)$
- (iii) $\frac{(x+2)^2}{(2x+1)(x-1)} > 0$
- (iv) $\frac{(x-\sqrt{2})(x+\sqrt{2})(x-3)^3}{(x+1)^2(x-4)} \leq 0$
- (v) $\frac{(x+2)(x-1)^2}{(x+1)(x-4)} \leq 0 \Rightarrow$
- D-2.** (i) $\frac{7x-5}{8x+3} - 4 > 0$
- (ii) $\frac{14x}{x+1} - \left(\frac{9x-30}{x-4}\right) < 0 \Rightarrow \frac{14x^2 - 56x - 9x^2 - 9x + 30x + 30}{(x+1)(x-4)} < 0$
 $\Rightarrow \frac{5x^2 - 35x + 30}{(x+1)(x+4)} < 0 \Rightarrow \frac{(x-6)(x-1)}{(x+1)(x+4)} < 0$
 $\Rightarrow x \in (-6, -1) \cup (1, 4)$
- (iii) $\frac{-12x^2 - 12}{(x+1)(x+2)(x+3)} \leq 0$
- $x \in (-3, -2) \cup (-1, \infty)$
- (iv) $\frac{x^2 + 2 + 2x^2 - 2}{x^2 - 1} < 0 \Rightarrow \frac{3x^2}{(x-1)(x+1)} < 0$
- $\Rightarrow x \in (-1, 0) \cup (0, 1)$





D-3. (i) $\frac{+ND-0+ND-0+}{-2 \quad \frac{3-\sqrt{5}}{2} \quad 1 \quad \frac{3+\sqrt{5}}{2}}$ $x \in \left(-2, \frac{3-\sqrt{5}}{2}\right] \cup \left[1, \frac{3+\sqrt{5}}{2}\right]$

(ii) $2x^2 - 3x - 459 > x^2 + 1$

$\Rightarrow x^2 - 3x - 460 > 0 \Rightarrow x \in (-\infty, -20) \cup (23, \infty)$

(iii) $x^2 - 5x + 12 > 3(x^2 - 4x + 5)$ (since $x^2 - 4x + 5 = (x-2)^2 + 1 > 0$)

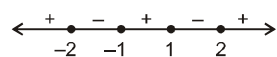
$\Rightarrow 2x^2 - 7x + 3 < 0 \Rightarrow 2x^2 - 6x - x + 3 < 0$

$\Rightarrow 2x(x-3) - 1(x-3) < 0 \Rightarrow x \in \left(\frac{1}{2}, 3\right)$

(iv) $x^4 + x^2 + 1$ is always positive सदैव धनात्मक

$\Rightarrow \frac{1}{(x-5)(x+1)} > 0$

D-4. (i) $(x^2 - 1)(x^2 - 4) \leq 0 \Rightarrow (x-1)(x+1)(x+2)(x-2) \leq 0$



$x \in [-2, -1] \cup [1, 2]$

(ii) $(x^2 - 9)(x^2 + 7) \leq 0 \Rightarrow (x-3)(x+3) \leq 0$

$x \in [-3, 3]$

D-5. $\frac{x-1-7x-14}{x+2} < 0$ and $\frac{x-1-x-2}{x+2} > 0$

$x \in (-\infty, -\frac{5}{2}) \cup (-2, \infty)$ and $x \in (-\infty, -2) \Rightarrow x \in (-\infty, -\frac{5}{2})$

(i) $x \in (-\infty, -\frac{5}{2})$ (ii) $x^2 \in (\frac{25}{4}, \infty)$ (iii) $\frac{1}{x} \in \left(-\frac{2}{5}, 0\right)$

D-6. $\frac{-ND+ND-0+0-}{\frac{-5-\sqrt{13}}{2} \quad \frac{-5+\sqrt{13}}{2} \quad 0 \quad 2}$

Number of positive integer satisfying the inequality equal to 2 (which are 1 and 2)

Section (E)

E-1. (i) $\log_{10}(\log_{10} 5 + 2\log_{10} 2) + (\log_{10} 2)^2 = (\log_{10} 5 + \log_{10} 2)^2 = (\log_{10} 10)^2 = 1$

(ii) $2^{\log_{\sqrt{5}} 5} + 7^{\log_3 9} + 5^{\log_2 8} = 2^2 + 7^2 - 5^3 = 53 - 125 = -72$

(iii) $\left(5^{\log_5 7} + \frac{1}{\left(\log_{10} \left(\frac{1}{0.1}\right)\right)}\right)^{1/3} = (7 + 1)^{1/3} = 2$

(iv) $\log_{3/4} \log_2 \left((8)^{1/2}\right)^{1/2} = \log_{3/4} \log_2 (2)^{3/4} = 1$

(v) $\left(\frac{1}{49}\right)^{1+\log_7^2} = (7^{-2})^{\log_7 7 + \log_7 2} = (7^{-2})^{\log_7^{14}} = 7^{\log_7 (14)^{-2}} = \frac{1}{196}$ & $5^{-\log_{1/5} 7} = 5^{\log_5 7} = 7$

$\therefore 7 + \frac{1}{196}$

(vi) $7^{\log_3 5} + 3^{\log_5 7} - 5^{\log_3 7} - 7^{\log_5 3} = 7^{\log_3 5} + 3^{\log_5 7} - 7^{\log_3 5} - 3^{\log_5 7}$

{using property $a^{\log_c b} = b^{\log_c a}$ } = 0



- E-2.** (i) $\sqrt{2} > 1$ & $\sqrt{3} > 1$, hence positive
 (ii) Number & base on opposite sides of unity, hence negative
 (iii) Number & base on same sides of unity, hence positive
 (iv) Number & base on same sides of unity, hence positive
 (v) Number & base on same sides of unity, hence positive
 (vi) Number & base on opposite sides of unity, hence negative
 (vii) Number & base on same sides of unity, hence positive
 (viii) Number & base on opposite sides of unity, hence negative
 (ix) Number & base on opposite sides of unity, hence negative

E-3. (i) $\log_{10}\left(\frac{3}{4}\right) = \log_{10}3 - 2\log_{10}2 = b - 2a$

(ii) $\log_{10}2 + 3\log_{10}3 = a + 3b$

(iii) $\frac{2\log_{10}3}{\log_{10}2} + \frac{3\log_{10}2}{\log_{10}3} = \frac{2b}{a} + \frac{3a}{b} = \frac{2b^2 + a^2}{ab}$

(iv) $2\left(\frac{4\log_{10}2 + 2\log_{10}3}{2\log_{10}3 + \log_{10}5}\right) = \frac{4(2a + b)}{1 - a + 2b}$

E-4. (i) $n = 75600$

Now $4\log_n2 + 3\log_n3 + 2\log_n5 + \log_n7 = \log_n(2^4 \cdot 3^3 \cdot 5^2 \cdot 7^1) = \log_n(75600) = 1$

(ii) $x = 4^{3^2} = 4^3 = 64$

$y = (2^4)^1 = 16, \quad z = 3^{2^1} = 9$
 $\text{sum} = 64 + 16 + 9 = 89$

(iii) $a_n = \frac{1}{\log_n 2002} = \log_{2002} n$

$b = a_2 + a_3 + a_4 + a_5 = \log_{2002} 2 + \log_{2002} 3 + \log_{2002} 4 + \log_{2002} 5 = \log_{2002} (2 \cdot 3 \cdot 4 \cdot 5) = \log_{2002} 120$

$c = a_{10} + a_{11} + a_{12} + a_{13} + a_{14} = \log_{2002} 10 + \log_{2002} 11 + \log_{2002} 12 + \log_{2002} 13 + \log_{2002} 14$
 $= \log_{2002} (10 \cdot 11 \cdot 12 \cdot 13 \cdot 14) = \log_{2002} 240240$

Now $b - c = \log_{2002} 120 - \log_{2002} 240240 = \log_{2002} 10 = a_1$

E-5. Let $\log_2 7 = \frac{p}{q}$, where p & q are coprime numbers.

$\Rightarrow 7 = 2^{p/q} \quad \Rightarrow 7^q = 2^p$

$\therefore 7^q$ is an odd number
 while 2^p is an even number

$\therefore$ this is not possible & $\log_2 7$ is an irrational number

E-6. $\frac{\log a}{b-c} = \frac{\log b}{c-a} = \frac{\log c}{a-b} = k$ (let)

$\log_a = k(b-c) \Rightarrow a \log a = ka(b-c)$

$\log_b = k(c-a) \Rightarrow b \log b = kb(c-a)$

$\log_c = k(a-b) \Rightarrow c \log c = kc(a-b)$

$\therefore \log a^a + \log b^b + \log c^c = k(ab - ac + bc - ab + ca - bc)$

$\Rightarrow \log a^a b^b c^c = 0 \quad \Rightarrow a^a b^b c^c = 1$

Section (F)

F-1. (i) $\log_x(4x-3) = 2 \Rightarrow 4x-3 = x^2 \Rightarrow x^2 - 4x + 3 = 0$
 $\Rightarrow x^2 - 4x + 3 = 0 \Rightarrow (x-1)(x-3) = 0 \Rightarrow x = 1, x = 3$
 But $4x-3 > 0 \Rightarrow x > \frac{3}{4}$ and $x > 0, x \neq 1$

Ans. $x = 3$

(ii) $\log_2 \log_3(x^2 - 1) = 0 \Rightarrow \log_3(x^2 - 1) = 2^0 = 1 \Rightarrow (x^2 - 1) = 3^1$

$\Rightarrow x^2 - 1 = 3 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$

both are satisfied





- (iii) $x^2 - 2x - 3 = 0 \Rightarrow (x-3)(x+1) = 0 \quad x = 3, x = -1$
but $x > 0$
Ans. $x = 3$
- (iv) $\log_4(\log_2 x) + \log_2(\log_4 x) = 2 \Rightarrow \frac{1}{2} \log_2(\log_2 x) + \log_2 \left[\frac{1}{2} \log_2 x \right] = 2$
 $\frac{1}{2} \log_2(\log_2 x) + \log_2(\log_2 x) = 2 \Rightarrow \frac{3}{2} \log_2(\log_2 x) = 3$
 $\log_2(\log_2 x) = 2 \Rightarrow \log_2 x = 4 \Rightarrow x = 2^4 = 16$
- (v) $\log_3 \left(\log_9 x + \frac{1}{2} + 9^x \right) = 2x \Rightarrow \log_9 x + \frac{1}{2} + 9^x = 3^{2x} \Rightarrow \log_9 x + \frac{1}{2} + 9^x = 9^x$
 $\Rightarrow \log_9 x = -\frac{1}{2} \Rightarrow x = 9^{-1/2} \Rightarrow x = \frac{1}{3}$
- (vi) $2 \log_4(4-x) = 4 - \log_2(-2-x)$
(i) $4-x > 0 \Rightarrow x < 4$
(ii) $-2-x > 0 \Rightarrow x < -2$
(iii) $\log_2(4-x) = 4 - \log_2(-2-x) \Rightarrow \log_2(4-x)(-2-x) = 4$
 $\Rightarrow (4-x)(-2-x) = 16 \Rightarrow -8 - 2x + x^2 = 16$
 $\Rightarrow x^2 - 2x - 24 = 0 \Rightarrow (x-6)(x+4) = 0$
 $x = 6$ (not possible), $x = -4$.
- (viii) $x^2 \log_{\sqrt{x}}(x^2-x) = 3 \log_{3^2}(2^2) = 2$
 $\Rightarrow (\sqrt{x})^{\log_{\sqrt{x}}(x^2-x)} = 2$
 $\Rightarrow x^2 - x = 2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x-2)(x+1) = 0$
 $\Rightarrow x = 2, x = -1$ but $x > 0$ and $x^2 - x = 0 \Rightarrow x(x-1) > 0$
 $x \in (-\infty, 0) \cup (1, \infty)$
- F-2.** (i) $t^2 - 2t - 5 = 0$
sum of roots = 2
 $\log_3 x_1 + \log_3 x_2 = 2 \Rightarrow \log_3(x_1 x_2) = 2 \Rightarrow x_1 x_2 = 3^2 = 9$
- (ii) $(2^x)^2 - 7(2^x) + 6 = 0$
 $\Rightarrow t^2 - 7t + 6 = 0$
 $t = 1, t = 6$
Roots $2^x = 1, 2^x = 6$
 $\Rightarrow x = 1, x = \log_2 6$
Product of roots = $(1)(\log_2 6) = \log_2 6$
- (iii) $x^{\log_{10} x + 2} = 10^{\log_{10} x + 2}$
 $x = 10$ or $\log_{10} x + 2 = 0 \Rightarrow x = 10^{-2} = \frac{1}{100}$
- (iv) $x^{\frac{\log x + 5}{3}} = 10^{5 + \log x} ; \left(\frac{\log x + 5}{3} \right) \log x = 5 + \log x$
 $\log^2 x + 2 \log x - 15 = 0 ; (\log x + 5)(\log x - 3) = 0$
 $\log x = -5, \log x = 3 ; x = 10^{-5}, x = 10^3.$

Section (G) :

- G-1.** (i) $\log_{\frac{5}{8}} \left(2x^2 - x - \frac{3}{8} \right) \geq 1 \Rightarrow 2x^2 - x - \frac{3}{8} \leq \frac{5}{8} \Rightarrow 16x^2 - 8x - 8 \leq 0$
 $\Rightarrow 2x^2 - x - 1 \leq 0 \Rightarrow (2x+1)(x-1) \leq 0$
 $x \in \left[-\frac{1}{2}, 1 \right] \dots\dots(i)$
also $2x^2 - x - \frac{3}{8} > 0 \Rightarrow 16x^2 - 8x - 3 > 0$





$$\Rightarrow 16x^2 - 12x + 4x - 3 > 0 \Rightarrow (4x - 3)(4x + 1) > 0$$

$$\Rightarrow x \in \left(-\infty, -\frac{1}{4}\right) \cup \left(\frac{3}{4}, 1\right] \quad \dots\dots(ii)$$

$$(i) \cap (ii) \Rightarrow x \in \left[\frac{-1}{2}, -\frac{1}{4}\right) \cup \left(\frac{3}{4}, 1\right]$$

$$(ii) \quad x^2 - 5x + 6 > 0 \Rightarrow (x - 3)(x - 2) > 0$$

$$\Rightarrow x \in (-\infty, 2) \cup (3, \infty) \quad \dots\dots(i)$$

$$\text{and } x^2 - 5x + 6 < \left(\frac{1}{2}\right)^{-1}$$

$$\Rightarrow x^2 - 5x + 4 < 0 \Rightarrow (x - 1)(x - 4) < 0 \Rightarrow x \in (1, 4) \quad \dots\dots(ii)$$

$$(i) \cap (ii) \Rightarrow x \in (1, 2) \cup (3, 4)$$

$$(iii) \quad \log_7 \left(\frac{2x - 6}{2x - 1} \right) > 0 \quad \Rightarrow \frac{2x - 6}{2x - 1} > 7^0 \quad \Rightarrow \frac{2x - 6}{2x - 1} - 1 > 0$$

$$\Rightarrow \frac{-5}{2x - 1} > 0 \quad \Rightarrow \frac{1}{2x - 1} < 0 \quad \Rightarrow x \in \left(-\infty, \frac{1}{2}\right)$$

$$\text{and } \frac{2x - 6}{2x - 1} > 0$$

$$\text{Ans. } \left(-\infty, \frac{1}{2}\right)$$

$$\Rightarrow \begin{array}{c} + \quad - \quad + \\ \hline \quad \quad 1/2 \quad \quad 3 \end{array}$$

$$(iv) \quad 2 - x < \frac{2}{x + 1} \Rightarrow \frac{2}{x + 1} + x - 2 > 0 \Rightarrow \frac{2 + x^2 - x - 2}{x + 1} > 0$$

$$\Rightarrow \frac{x(x - 1)}{x + 1} > 0 \Rightarrow \begin{array}{c} - \quad + \quad - \quad + \\ \hline -1 \quad 0 \quad 1 \end{array}$$

$$\text{and } 2 - x > 0 \Rightarrow x < 2 \quad \text{and} \quad \frac{2}{x + 1} > 0 \Rightarrow x > -1$$

$$\text{Ans. } (-1, 0) \cup (1, 2)$$

$$(v) \quad 2^2 \cdot 2^x - 4^x \leq \left(\frac{1}{3}\right)^{-2} \Rightarrow 4 \cdot 2^x - (2^x)^2 \leq 9$$

$$\text{Let } 2^x = t \Rightarrow t^2 - 4t + 9 \geq 0 \quad \text{always true}$$

$$2^{x+2} - 4^x > 0 \Rightarrow 2^x \cdot (4 - 2^x) > 0$$

$$4 - 2^x > 0 \Rightarrow 2^x < 2^2$$

$$\Rightarrow x < 2 \Rightarrow x \in (-\infty, 2)$$

$$(vi) \quad \log_x(4x - 3) \geq 2$$

$$\text{Case-I } 0 < x < 1 \quad \text{and } 4x - 3 > 0 \Rightarrow x > \frac{3}{4}$$

$$\text{then } 4x - 3 \leq x^2 \Rightarrow x^2 - 4x + 3 \geq 0 \Rightarrow (x - 3)(x - 1) \geq 0$$

$$\Rightarrow x \in \left(\frac{3}{4}, 1\right)$$

$$\text{Case -II } x > 1, 4x - 3 > 0 \Rightarrow x > \frac{3}{4}$$

$$\text{then } 4x - 3 \geq x^2 \Rightarrow x^2 - 4x + 3 \leq 0$$

$$\Rightarrow (x - 3)(x - 1) \leq 0 \Rightarrow x \in (1, 3]$$

$$\text{Ans. } x \in \left(\frac{3}{4}, 1\right) \cup (1, 3]$$





G-2. $\log_{\frac{1}{5}} \frac{4x+6}{x} \geq 0$

$$\frac{4x+6}{x} > 0 \Rightarrow x \in \left(-\infty, -\frac{3}{2}\right) \cup (0, \infty) \quad \dots(i)$$

$$\& \quad \frac{4x+6}{x} \leq 1 \Rightarrow \frac{x+2}{x} \leq 0 \Rightarrow x \in (-\infty, -2] \cup (0, \infty) \quad \dots(ii)$$

$$(i) \cap (ii) \Rightarrow x \in \left[-2, -\frac{3}{2}\right)$$

G-3 (i) $-2 \leq \log_5 x \leq 1$

$$x \in [(0.5)^1, (0.5)^{-2}] \Rightarrow x \in \left[\frac{1}{2}, 4\right]$$

(ii) $15^x - 25 \cdot 3^x - 9.5^x + 225 > 0 \quad (5^x - 25)(3^x - 9) \geq 0 \Rightarrow x \in \mathbb{R}$

(iii) $8 \left(\frac{3^{x-2}}{3^x - 2^x} \right) > 1 + \left(\frac{2}{3} \right)^x \Rightarrow \frac{8}{9} \left(\frac{\left(\frac{3}{2}\right)^x}{\left(\frac{3}{2}\right)^x - 1} \right) > 1 + \left(\frac{2}{3} \right)^x$

Let $\left(\frac{3}{2}\right)^x = t$ then $\frac{8t}{9(t-1)} > \frac{t+1}{t} \Rightarrow \frac{8t^2 - 9(t^2 - 1)}{9t(t-1)} > 0$

$$\Rightarrow \frac{(t-3)(t+3)}{t(t-1)} < 0 \Rightarrow t \in (-3, 0) \cup (1, 3)$$

$$\Rightarrow \left(\frac{3}{2}\right)^x \in (1, 3) \Rightarrow x \in (0, \log_{\frac{3}{2}} 3)$$

G-4. (i) $\log_x(4x-3) \geq 2$

Case -I : When $x > 1 \Rightarrow 4x-3 \geq x^2 \Rightarrow x^2 - 4x + 3 \leq 0$
 $\Rightarrow (x-1)(x-3) \leq 0$
 $x \in (1, 3] \quad \dots(i)$

Case-II : When $0 < x < 1$ and $4x-3 > 0 \Rightarrow 4x-3 \leq x^2$
 $\Rightarrow (x-1)(x-3) \geq 0$

$x > 3/4 \Rightarrow x \in \left(\frac{3}{4}, 1\right) \quad \dots(ii) \quad \text{Ans. } (i) \cup (ii)$

(ii) $\log_{3x^2+1} 2 < \frac{1}{2}$

$$\begin{aligned} 3x^2 + 1 > 1 &\Rightarrow x^2 > 0 \Rightarrow x \in \mathbb{R} - \{0\} \\ 2 < (3x^2 + 1)^{1/2} &\Rightarrow 3x^2 + 1 > 4 \Rightarrow (x-1)(x+1) > 0 \\ \Rightarrow x &\in (-\infty, -1) \cup (1, \infty) \end{aligned}$$

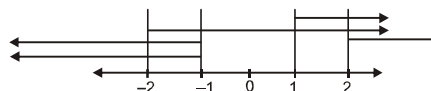
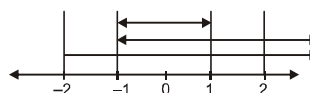
(iii) $\log_{x^2} (x+2) < 1 \quad x+2 > 0 \quad x > -2$

Case -I : when $0 < x^2 < 1 \Rightarrow x \in (-1, 0) \cup (0, 1)$
then $x+2 > x^2 \Rightarrow x^2 - x - 2 < 0$

$$x \in (-1, 1) - \{0\}$$

Case -II : $x^2 > 1 \Rightarrow |x| > 1$
 $x+2 < x^2 \Rightarrow x^2 - x - 2 > 0$
 $x \in (-2, -1) \cup (2, \infty)$

Hence, $x \in (-2, -1) \cup (-1, 0) \cup (0, 1) \cup (2, \infty)$





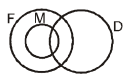
PART - II

Section (A) :

- A-1.** Since, intelligency is not defined for students in a class so set of intelligent students in a class is not well defined collection.
- A-2.** $x^2 = 16 \Rightarrow x = \pm 4$
 $2x = 6 \Rightarrow x = 3$
 No common value of x
- A-3.** $A = \{-2, -1, 0, 1, 2\}$
 No. of subsets $= 2^n = 2^5 = 32$
- A-4.** Obvious
- A-5.** $P(A) = \{\phi, \{7\}, \{10\}, \{11\}, \{7, 10\}, \{7, 11\}, \{10, 11\}, \{7, 10, 11\}\}$
 Number of subsets $= 2^n = 2^8 = 256$
- A-6.** Between any two real numbers there lie infinitely many real numbers.

Section (B) :

- B-1.** $A = \{x : x \in \mathbb{R}, -1 < x < 1\}$
 $B = \{x : x \in \mathbb{R} : x \leq 0 \text{ or } x \geq 2\}$
 $\therefore A \cup B = \mathbb{R} - D$, where $D = \{x : x \in \mathbb{R}, 1 \leq x < 2\}$
- B-2.** $A \cap B = \{3, 4, 10\}$
 $A \cap C = \{4\}$
 $(A \cap B) \cup (A \cap C) = \{3, 4, 10\}$
- B-3.** Obviously $A - (B \cup C)$
- B-4.** $B' = U - B = \{1, 2, 3, 4, 5, 8, 9, 10\}$
 $A \cap B' = \{1, 2, 5\} = A$
- B-5.** $A = \{5, 9, 13, 17, 21\}$ and $B = \{3, 6, 9, 12, 15, 18, 21, 24\}$
 $A - B = \{5, 13, 17\}$
 $A - (A - B) = \{9, 21\}$
- B-6.** Let $A \cup B = A \cap B$
 Now, $x \in A \Rightarrow x \in A \cup B$ ($\because A \subset A \cup B$)
 $\Rightarrow x \in A \cap B$ ($\because A \cup B = A \cap B$)
 $\Rightarrow x \in B$
 Similarly, $x \in B$ implies $x \in A$ $\therefore A = B$
 Conversely, let $A = B$
 $\therefore A \cup B = A \cup A = A = A \cap A = A \cap B$
 $\therefore A \cup B = A \cap B$
- B-7.** 1. $(N \cup B) \cap Z = (N \cap Z) \cup (B \cap Z) = N \cup (B \cap Z)$
 2. $A = \{3, 6, 9, 12, 15, 18, 21, 24\}$

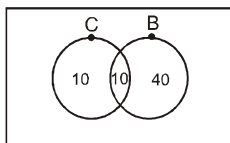


- B-8.** $M \equiv \text{Mother}; F \equiv \text{Female}; D = \text{Doctor}$

Section (C) :

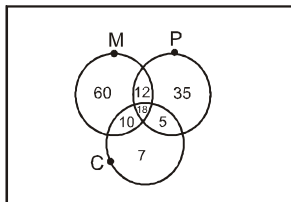
- C-1.** (i) $A \cup B \geq A \cup B$ (ii) $A \cap B \leq A \cup B$ (iii) $A \cap B = A \cup B$ not always





C-2.

$$P = 10 + 10 + 40 = 60\%$$



C-3.

Number of students offered maths alone = 60

$$n(M) = 100$$

$$n(P) = 70$$

$$n(C) = 40$$

$$n(M \cap P) = 30$$

$$n(M \cap P) = 28$$

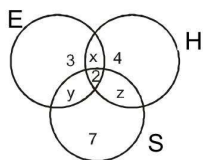
$$n(P \cap C) = 23$$

$$n(M \cap P \cap C) = 18$$

C-4.

$$x + y = 10; \quad x + z = 9; \quad y + z = 11 \Rightarrow x + y + z = 15$$

$$x = 4, y = 6, z = 5$$

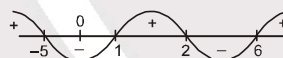


Section (D) :

$$D-1. \quad \frac{x^2(x^2 - 3x + 2)}{x^2 - x - 30} \geq 0 \Rightarrow \frac{x^2(x-1)(x-2)}{(x+5)(x-6)} \geq 0$$

$$x \neq -5, 6$$

$$x \in (-\infty, -5) \cup [1, 2] \cup (6, \infty) \cup \{0\}$$



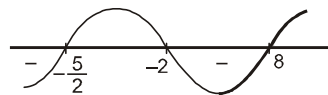
D-2.

$$x \in (-\infty, -9) \cup (-9, -3) \cup [-1, 0) \cup (0, 2) \cup [4, 6)$$

so +ve integral solution

$$D-3. \quad \frac{x^2 - 1}{2x + 5} - 3 < 0 \Rightarrow \frac{x^2 - 1 - 6x - 15}{2x + 5} < 0 \Rightarrow \frac{x^2 - 6x - 16}{2x + 5} < 0 \Rightarrow \frac{x^2 - 8x + 2x - 16}{\left(x + \frac{5}{2}\right)} < 0$$

$$\Rightarrow \frac{x(x-8) + 2(x-8)}{x + \frac{5}{2}} < 0 \Rightarrow \frac{(x-8)(x-2)}{x + \frac{5}{2}} < 0$$



$$x \in (-\infty, -5/2) \cup (-2, 8)$$

D-4.

$$\frac{x^2 - 1}{x + 3} \geq 0$$

$$x^2 - 5x + 2 \leq 0.$$

$$\Rightarrow x \in (-3, -1] \cup [1, \infty)$$

$$\Rightarrow x \in \left[\frac{5 - \sqrt{17}}{2}, \frac{5 + \sqrt{17}}{2} \right]$$

$$\text{so the common solution is } x \in \left[1, \frac{5 + \sqrt{17}}{2} \right]$$





D-5. $2x - 1 \leq x^2 + 3 \leq x - 1$
 $x^2 + 3 \leq x - 1 \Rightarrow x^2 - x + 4 \leq 0$ which is not true for $x \in \mathbb{R}$

D-6. $x^2 + 9 < (x + 3)^2 < 8x + 25$
 $\Rightarrow (x + 3)^2 > x^2 + 9 \Rightarrow x > 0$ (i)
 and $(x + 3)^2 < 8x + 25$
 $\Rightarrow x^2 - 2x - 16 < 0$
 $\Rightarrow x \in (1 - \sqrt{17}, 1 + \sqrt{17})$ (ii)
 $(i) \cap (ii) \Rightarrow x \in (0, 1 + \sqrt{17})$
 Number of integers = 5

D-7. $\frac{2}{x^2 - x + 1} - \frac{1}{x + 1} - \frac{2x - 1}{x^3 + 1} \geq 0$
 $\frac{2x + 2 - x^2 + x - 1}{x^3 + 1} - \frac{(2x - 1)}{x^2 + 1} \geq 0 = \frac{3x + 1 - x^2 - 2x + 1}{x^3 + 1} \geq 0 = \frac{-x^2 + x + 2}{x^3 + 1} \geq 0 = \frac{x^2 - x - 2}{x^3 + 1} \leq 0$
 $= \frac{(x + 1)(x - 2)}{(x + 1)(x^2 - x + 1)} \leq 0 \Rightarrow \frac{(x - 2)}{(x^2 - x + 1)} \leq 0$ $\xrightarrow{(-\infty, -1) \cup (-1, 2] \quad 2}$
 required value of x , $\{0, 1, 2\}$

Section (E) :

E-1. $a^4 b^5 = 1 \Rightarrow \log \text{ w.r.t. } a \rightarrow 4 + 5\log_a b = \log_a 1 \Rightarrow \log_a b = -\frac{4}{5}$

Now $\log_a(a^5 b^4) = 5 + 4\log_a b = 5 + 4\left(-\frac{4}{5}\right) = \frac{25 - 16}{5} = \frac{9}{5}$

E-2. $\frac{1}{1 + \log_b a + \log_b c} + \frac{1}{1 + \log_c a + \log_c b} + \frac{1}{1 + \log_a b + \log_a c} = \frac{1}{\log_b abc} + \frac{1}{\log_c abc} + \frac{1}{\log_a abc}$
 $= \log_{abc} b + \log_{abc} c + \log_{abc} a = \log_{abc} abc = 1$

E-3. $\frac{1}{\log_{\sqrt{bc}} abc} + \frac{1}{\log_{\sqrt{ca}} abc} + \frac{1}{\log_{\sqrt{ab}} abc} = \log_{abc} \sqrt{bc} + \log_{abc} \sqrt{ca} + \log_{abc} \sqrt{ab} = \log_{abc} abc = 1$

E-4. Obvious

E-5. $y = \frac{2^{\log_{2^{1/4}} a} - 3^{\log_{27}(a^2 + 1)^3} - 2a}{7^{4\log_{49} a} - a - 1} \Rightarrow 2^{\log_{2^{1/4}} a} = 2^{4\log_2 a} = a^4$
 $3^{\log_{27}(a^2 + 1)^3} = 3^{\log_3(a^2 + 1)} = a^2 + 1 \Rightarrow 7^{4\log_{49} a} = 7^{2\log_7 a} = a^2$
 $\therefore y = \frac{a^4 - (a^2 + 1 + 2a)}{a^2 - a - 1} = \frac{a^4 - (a + 1)^2}{a^2 - a - 1} = a^2 + a + 1$

E-6. $\log_a(ab) = x \Rightarrow 1 + \log_a b = x \Rightarrow \log_a b = x - 1 \Rightarrow \log_b a = \frac{1}{x - 1}$

Now $\log_b(ab) = 1 + \log_b a = 1 + \frac{1}{x - 1} = \frac{x - 1 + 1}{x - 1} = \frac{x}{x - 1}$

E-7. $\log_p(\log_q(\log_r x)) = 0 \Rightarrow \log_q(\log_r x) = b \Rightarrow \log_r x = q \Rightarrow x = r^q$ (i)

and $\log_q(\log_r(\log_p x)) = 0$

$\Rightarrow \log_r(\log_p x) = 1 \Rightarrow \log_p x = r \Rightarrow x = p^r$ (ii)

from (i) and (ii) $p^r = r^q$

$\Rightarrow p = r^{q/r}$

E-8. $\log_{10} \pi$ is quantity lie between 0 to 1.

E-9. $\log_{10}(\log_2 3 \cdot \log_3 4 \cdot \log_4 5 \dots \log_{1023} 1024) = \log_{10}(\log_2 1024) = \log_{10}(\log_2 2^{10}) = \log_{10}(10) = 1$



**Section (F) :**

$$\begin{aligned} \text{F-1. } 2\log_{10}x - \log_{10}(2x - 75) &= 2 \Rightarrow \frac{x^2}{2x - 75} = 10^2 = 100 \\ &\Rightarrow x^2 - 200x + 7500 = 0 \Rightarrow x = 50, x = 150 \end{aligned}$$

$$\begin{aligned} \text{F-2. } \text{sum} &= 200 \\ \text{Take } \log_5 x &= t \\ \text{on solving we get } x &= 1/625 \text{ \& } x = 5. \\ \log_5 x &= t \\ x &= 1/625 \text{ \& } x = 5. \end{aligned}$$

$$\begin{aligned} \text{F-3. } \text{Take } \log_x 5 &= t \text{ on solving we get } x = \sqrt[3]{5} \text{ and } 5. \\ \log_x 5 &= t \end{aligned}$$

$$\text{F-4. } \log_p \log_p (p)^{\frac{1}{p^n}} = \log_p \left(\frac{1}{p} \right)^n = -\log_p p^n = -n \quad \text{independent of } p.$$

$$\begin{aligned} \text{F-5. } \log_x \log_{18} (\sqrt{2} + 2\sqrt{2}) &= \frac{1}{3} \Rightarrow \log_x \log_{18} (\sqrt{18}) = \frac{1}{3} \Rightarrow \log_x \frac{1}{2} = \frac{1}{3} \\ \Rightarrow x^{1/3} &= \frac{1}{2} \Rightarrow x = \frac{1}{8} \Rightarrow 1000x = 125. \end{aligned}$$

$$\begin{aligned} \text{F-6. } \sqrt{\log_{10}(-x)} &= \log_{10}|x| \Rightarrow -x > 0 \Rightarrow x < 0 \\ \therefore |x| &= -x \Rightarrow \sqrt{\log_{10}(-x)} = \log_{10}(-x) \\ \log_{10}(-x) (\log_{10}(-x) - 1) &= 0 \\ \log_{10}(-x) &= 0 \quad \log_{10}(-x) = 1 \\ \Rightarrow -x &= 1 \quad -x = 10 \\ x &= -1 \quad x = -10. \end{aligned}$$

F-7. Clearly Domain is $x > 0$ and $x \neq 1$

Section (G) :

$$\begin{aligned} \text{G-1. } \log_{\sin \frac{\pi}{3}} (x^2 - 3x + 2) &\geq 2 \Rightarrow x^2 - 3x + 2 \leq \left(\frac{\sqrt{3}}{2} \right)^2 \Rightarrow 4x^2 - 12x + 8 \leq 3 \\ \Rightarrow 4x^2 - 12x + 5 &\leq 0 \Rightarrow (2x - 5)(2x - 1) \leq 0 \Rightarrow x \in \left[\frac{1}{2}, \frac{5}{2} \right] \\ \text{But domain } x^2 - 3x + 2 &> 0 \Rightarrow (x - 1)(x - 2) > 0 \Rightarrow x \in (-\infty, 1) \cup (2, \infty) \\ \text{Hence } x &\in \left[\frac{1}{2}, 1 \right) \cup \left(2, \frac{5}{2} \right] \end{aligned}$$

$$\begin{aligned} \text{G-2. } \log_{0.3} (x - 1) &< \log_{0.09} (x - 1) \quad ; \quad \log_{0.3} (x - 1) < \frac{\log_{0.3} (x - 1)}{2} \\ \Rightarrow \log_{0.3} (x - 1) &< 0 \Rightarrow x - 1 > 1 \Rightarrow x > 2 \end{aligned}$$

$$\begin{aligned} \text{G-3. } 2 - \log_2 (x^2 + 3x) &\geq 0 \Rightarrow \log_2 (x^2 + 3x) \leq 2 \\ x^2 + 3x &> 0 \Rightarrow x \in (-\infty, -3) \cup (0, \infty) \quad \dots(i) \\ \text{and } x^2 + 3x &\leq 4 \Rightarrow x \in [-4, 1] \quad \dots(ii) \\ \Rightarrow (x - 1)(x + 4) &\leq 0 \Rightarrow x \in [-4, 1] \quad \dots(ii) \\ (i) \cap (ii) &\Rightarrow x \in [-4, -3) \cup (0, 1] \end{aligned}$$

$$\begin{aligned} \text{G-4. } \log_{0.5} \log_5 (x^2 - 4) &> \log_{0.5} 1; \quad \log_{0.5} \log_5 (x^2 - 4) > 0 \\ \Rightarrow x^2 - 4 &> 0 \Rightarrow x \in (-\infty, -2) \cup (2, \infty) \quad \dots(i) \\ \log_5 (x^2 - 4) &> 0 \Rightarrow x^2 - 5 > 0 \\ \Rightarrow x &\in (-\infty, -\sqrt{5}) \cup (\sqrt{5}, \infty) \quad \dots(ii) \\ \log_5 (x^2 - 4) &< 1 \\ \Rightarrow x^2 - 9 &< 0 \Rightarrow x \in (-3, 3) \quad \dots(iii) \\ (i) \cap (ii) \cap (iii) &\Rightarrow x \in (-3, \sqrt{5}) \cup (\sqrt{5}, 3) \end{aligned}$$





G-5. $\left(\frac{1}{2}\right)^{x^2-2x} < \left(\frac{1}{2}\right)^2 \Rightarrow x^2-2x > 2 \Rightarrow x^2-2x-2 > 0 \Rightarrow x \in (-\infty, 2-\sqrt{3}) \cup (2+\sqrt{3}, \infty)$

G-6. $\log_2(4^x - 2 \cdot 2^x + 17) > 5$
 $4^x - 2 \cdot 2^x + 17 > 0$
 $(2^x)^2 - 2 \cdot 2^x + 17 > 0 \Rightarrow \forall x \in \mathbb{R} \quad \text{and} \quad 4^x - 2 \cdot 2^x + 17 > 32$
 $\Rightarrow (2^x)^2 - 2 \cdot 2^x - 15 > 0 \Rightarrow (2^x + 3)(2^x - 5) > 0 \Rightarrow 2^x < -3 \text{ or } 2^x > 5$
 $\Rightarrow x \in \phi \quad \text{or} \quad x > \log_2 5 \Rightarrow x \in (\log_2 5, \infty)$

G-7. $\log_{1-x}(x-2) \geq -1$
 $x > 2$ (1)
 (i) When $0 < 1-x < 1 \Rightarrow 0 < x < 1$. So no common range comes out.
 (ii) When $1-x > 1 \Rightarrow x < 0$ but $x > 2$
 here, also no common range comes out. , hence no solution. Finally, no solution

PART - III

- (A) The set $\{3^{2n} - 8n - 1 : n \in \mathbb{N}\}$ contains 0 and every element of this set is a multiple of 64.
 (B) $2^{3n} - 1$ is always divisible by 7.
 (C) $3^{2n} - 1$ is always divisible by 8.
 (D) $2^{2n} - 7n - 1$ is always divisible by 49 and $2^{3n} - 7n - 1 = 0$ for $n = 1$.
- (A) $a = 3 \quad ((\sqrt{7} + 1) - (\sqrt{7} - 1))$
 $a = 3(2) = 6$
 $b = \sqrt{(42)(30) + 36} = 6\sqrt{7 \times 5 + 1} = 6 \times 6 = 36$
 $\log_a b = \log_6 36 = 2$

(B) $a = (\sqrt{3} + 1) - (\sqrt{3} - 1) = 2$
 $b = (3 + \sqrt{2}) - (3 - \sqrt{2}) = 2\sqrt{2}$
 $\log_a b = \log_2(2\sqrt{2}) = \log_2(2^{3/2}) = 3/2$

(C) $a\sqrt{3+2\sqrt{2}} = (\sqrt{2}+1) = , b = \sqrt{3-2\sqrt{2}} = (\sqrt{2}-1) = \frac{1}{\sqrt{2}+1}$
 $\log_a b = \log_{\sqrt{2}+1}(\sqrt{2}+1)^{-1} = -1$

(D) $a = \sqrt{7+\sqrt{7^2-1}} = \sqrt{7+\sqrt{48}} = \sqrt{7+4\sqrt{3}} = (2+\sqrt{3})$
 $b = 2-\sqrt{3} = \frac{1}{2+\sqrt{3}} = (2+\sqrt{3})^{-1}$ Now $\log_a b = \log_{2+\sqrt{3}}(2+\sqrt{3})^{-1} = -1$

(E) First 20 prime numbers are 2, 3, 5, 7, 11, 13, 17, 19,
 zero are formed by multiple of 2 and 5 that is one times 2 and one times 5 there fore one zero at end of product of first 20 prime numbers.

(F) $2^{2x} - 3^{2y} = 55, x, y \in \mathbb{I} \Rightarrow 4^x - 9^y = 55$
 only $x = 3, y = 1$ satisfy
 There fore number of solution is one set (x, y) g.e. (3, 1)
- (A) $x = 0.363636.....$
 $100x = 36.363636.....$
 $\hline 99x = 36$
 $\Rightarrow x = \frac{36}{99} = \frac{4}{11}$ sum of numerator and denominator is $4 + 11 = 15$

(C) $\frac{1}{\log_a 8} + \frac{1}{\log_b 8} = \frac{1}{\log_a 8 \cdot \log_b 8} \Rightarrow \log_b 8 + \log_a 8 = 1$ (given $\log_a b = 3$)
 $\Rightarrow 4\log_b 8 = 1 \quad \log_b 8 = 3\log_b a \quad \dots\dots(1)$
 $\Rightarrow \log_b 8 = 1/4 \quad \log_a 8 = 3\log_b 8$
 $8 = b^{1/4} \Rightarrow b = 8^4$
 $\log_8(8^4) = 3\log_8 a \Rightarrow \log_8 a = \frac{4}{3} \Rightarrow a = (8^{4/3}) = (2^3) = 2^4 = 16$





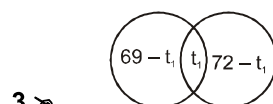
4. (A) $\text{Antilog}_{27}(0. \bar{6}) = x \Rightarrow 0. \bar{6} = \log_{27} x = \frac{2}{3} \Rightarrow x = (27)^{2/3} = (3^3)^{2/3} = 9$
- (B) Since $2^{10} < 2008 < 2^{11} \Rightarrow \log_2(2^{10}) < \log_2 2008 < \log_2(2^{11}) \Rightarrow 10 < \log_2 2008 < 11$
- (C) $\log_e 2 \cdot \log_b 625 = \log_{10} 16 \cdot \log_e 10 \Rightarrow \log_e 2 \cdot \log_b 265 = \log_e 16 \Rightarrow \log_b 625 = \log_2 16 = 4$
 $625 = b^4$
- (D) $x = \left(\frac{5}{6}\right)^{100} = \left(\frac{10}{2 \times 6}\right)^{100} = \left(\frac{10}{2^2 \cdot 3}\right)^{100}$
 $\log_{10} x = 10(1 - 2\log 2 - \log 3) = 100(1 - 2(0.3010) - 0.4771) = 100(-0.0791) = 7.91$

EXERCISE-2

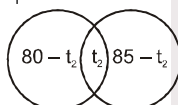
PART - I

1. $A_1 \cup A_2 \cup A_3$ is the smallest element containing subset of all we set A_1, A_2 and A_3

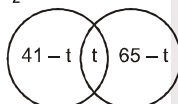
2. 1. $((A \cap B) \cup C)' \cap B'$
 $= (A \cap B) \cup C \cup B$
 $= (A \cap B) \cup B \cup C$
 $= B \cup C \neq B \cap C$
2. $(A' \cap B') \cap (A \cup B \cup C')$
 $= (A \cup B)' \cap ((A \cup B) \cup C')$
 $= \phi \cup ((A \cup B)' \cap C')$
 $= ((A \cup B) \cup C)'$
 $= (A \cup (B \cup C))'$



3. $70 + 72 - t_1 = 100$
 $t_1 = 42\% \Rightarrow \text{min. in } P \cap C = 42\%$



$t_2 = 85\% - 20\% = 65\% \Rightarrow \text{min. } M \cap E = 65\%$



$t = 42 - 35 = 7\%$
 $\text{min. in } ((P \cap C) \cap (M \cap E)) = 7\%$

4. $X \cap (Y \cup X)' = X \cap (Y' \cap X') = X \cap X' \cap Y' = \phi$
 $\Rightarrow$ Statement - 1 true.
 $X \Delta Y = (X \sim Y) \cup (Y \sim X) = (X \cup Y) \sim (X \cap Y) \Rightarrow \text{number of element in } X \Delta Y = m - n$
 $\Rightarrow$ Statement-2 is true but does explain statement-1

5. $\frac{6x^2 - 5x - 3}{x^2 - 2x + 6} - 4 \leq 0 \Rightarrow \frac{2x^2 + 3x - 27}{x^2 - 2x + 6} \leq 0$
denominator $x^2 - 2x + 6 > 0 \quad \forall x \in \mathbb{R} (\because D < 0)$
then $2x^2 + 3x - 27 \leq 0 \Rightarrow (2x + 9)(x - 3) \leq 0$
 $-\frac{9}{2} \leq x \leq 3 \Rightarrow 0 \leq x^2 \leq \frac{81}{4}$
 $(4x^2)_{\max} = 4\left(-\frac{9}{2}\right)^2 = 81 \Rightarrow (4x^2)_{\min} = 4(0) = 0$

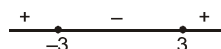
6. $x^2 - 16 \geq 0$
 $\therefore (x - 4)(x + 4) \geq 0$





$$\therefore x \in (-\infty, -4] \cup [4, \infty) \quad \dots\dots\dots(1)$$

$$\text{Now } \frac{(x^2 + 2)(\sqrt{x^2 - 16})}{(x^4 + 2)(x - 3)(x + 3)} \leq 0$$



$$x \in (-3, 3) \quad \dots\dots\dots(2)$$

$$\text{By (1) and (2) } x \in \{-4, 4\}$$

$$7. \quad b = a^2, c = b^2, \frac{c}{a} = 3^3 \Rightarrow c = 27a$$

$$\Rightarrow b^2 = 27a$$

$$\Rightarrow a^4 = 27a$$

$$\Rightarrow a = 3, a > 0$$

$$c = 81, b = 9$$

$$\therefore a + b + c = 3 + 9 + 81 = 93$$

$$8. \quad x = (-\log_3 5)(\log_{5^3}(7^3))(\log_{7^2}(3^6))$$

$$x = -\frac{3}{3} \cdot \frac{6}{2} \log_3 5 \cdot \log_5 7 \cdot \log_7 3 \quad \text{and और } y = 25^{(3 \log_{17^2} 11)(\log_{28} 17^{1/2})(\log_{(11)^3} (28)^2)}$$

$$y = 25^{\frac{3}{2} \cdot \frac{1}{2} \cdot \frac{2}{3} (\log_{17} 11) \cdot (\log_{28} 17) \cdot (\log_{11} 28)}$$

$$y = 5$$

$$\therefore x^2 + y^2 = (-3)^2 + 5^2 = 34$$

$$9. \quad \frac{(x+1)+3x-x\log_2 8}{(x-1)(1)} = \frac{(x+2)+3x-3x}{(x-1)} = \frac{x+1}{x-1}$$

$$10. \quad a^{(\log_3 7)^2} = (a^{\log_3 7})^{\log_3 7} = 27^{\log_3 7} = 27^{\log_3 7} = 7^3 = 343$$

$$b^{(\log_7 11)^2} = (b^{\log_7 11})^{\log_7 11} = 49^{\log_7 11} = 11^{\log_7 49} = 121$$

$$c^{(\log_{11} 25)^2} = (c^{\log_{11} 25})^{\log_{11} 25} = (\sqrt{11})^{\log_{11} 25} = 25^{\log_{11} \sqrt{11}} = 5$$

$$\text{hence the sum is } 343 + 121 + 5 = 469$$

$$11. \quad \text{For } 0 < x < y < 1$$

$$f(x) = x(\alpha - x) = -x^2 + \alpha x \quad f(x) < f(1)$$

so $f(x)$ should be increasing in $(0, 1)$ and one root is 0 so vertex should be ≥ 1

$$\Rightarrow \frac{-\alpha}{-2} \geq 1, \alpha \geq 2$$

$$12. \quad \frac{\sqrt{(x-8)(2-x)}}{\log_{0.3} \left(\frac{10}{7} (\log_2 5 - 1) \right)} \geq 0$$

For $\sqrt{(x-8)(2-x)}$ to be defined

$$(i) \quad (x-8)(2-x) \geq 0$$

$$(x-2)(x-8) \leq 0 \Rightarrow 2 \leq x \leq 8$$

$$\text{Now Let say } y = \log_{0.3} \frac{10}{7} (\log_2 5 - \log_2 2) = \log_{0.3} \frac{10}{7} (\log_2 5/2)$$

$$\text{Let } y < 0 \quad (\text{assume}) \text{ then } \log_{0.3} \frac{10}{7} (\log_2 5/2) < 0$$

$$\Rightarrow \frac{10}{7} \log_2 5/2 > 1 \Rightarrow \log_2 5/2 > \frac{7}{10} \Rightarrow \frac{5}{2} > 2^{(7/10)} \text{ which is true}$$

$$\text{So } y < 0$$



so denominator is -ve and numerator is +ve, so inequality is not satisfied,

$$\text{thus } \sqrt{(x-8)(2-x)} = 0$$

$$x = 2, 8 \quad \dots(i)$$

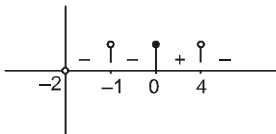
Now $2^{x-3} > 31$

$$\Rightarrow (x-3) > \log_2 31 \Rightarrow x > 3 + \log_2 2^{4.9} \text{ (approx)}$$

$$\Rightarrow x > 7.9 \Rightarrow x = 8$$

13. ~~A~~

$$\frac{4^x \left(\left(\frac{3}{4} \right)^x - 1 \right) \ln(x+2)}{(x-4)(x+1)} \leq 0$$



14.

$$\sqrt{\log_4 \{ \log_3 \{ \log_2 (x^2 - 2x + a) \} \}}$$

for defined $\log_4 \log_3 \log_2 (x^2 - 2x + a) \geq 0$

$$\Rightarrow \log_3 \log_2 (x^2 - 2x + a) \geq 1 \Rightarrow \log_2 (x^2 - 2x + a) \geq 3$$

$$\Rightarrow x^2 - 2x + a \geq 8 \Rightarrow x^2 - 2x + (a - 8) \geq 0$$

$$\Rightarrow D \leq 0$$

$$4 - 4(a - 8) \leq 0 \Rightarrow 1 - a + 8 \leq 0$$

$$\Rightarrow a \geq 9$$

15.

domain $x > -\frac{3}{2}$

$$\log_{2x+3} (6x^2 + 23x + 21) = 4 - \log_{(3x+7)} (4x^2 + 12x + 9)$$

$$\log_{(2x+3)} (2x+3) (3x+7) = 4 - \log_{3x+7} (2x+3)^2$$

$$1 + \log_{2x+3} (3x+7) = 4 - 2 \log_{3x+7} (2x+3)$$

$$\log_{2x+3} (3x+7) = y$$

$$y + \frac{2}{y} - 3 = 0 \Rightarrow y = 1 \text{ or } y = 2 \Rightarrow x = -2, -\frac{1}{4} - 4$$

$$x \neq -2, -4 \text{ so } x = -\frac{1}{4}$$

PART - II

1. ~~A~~

$$n(A \cup B) = 280$$

$$\text{Now } n(A' \cap B') = n(A \cup B)' = 2009 - n(A \cup B) = 2009 - 280 = 1729 = 12^3 + 1^3 = 10^3 + 9^3$$

2. ~~A~~

$$n(A - B) = 1681 - 1075 = 606 = 4 + 2 \times 301 = 4 + 2 \times 7 \times 43 = (2) 2 + 2 \times 7 \times 43$$

3.

For $A \cap B$

$$x^3 + (x-1)^3 = 1 \Rightarrow x^3 + x^3 - 3x^2 + 3x - 1 = 1 \Rightarrow 2x^3 - 3x^2 + 3x - 2 = 0 \Rightarrow (x-1)(2x^2 - x + 2) = 0$$

$$\Rightarrow x = 1 \Rightarrow y = 0 \Rightarrow (x, y) = (1, 0)$$

For $A \cap C$

$$x^3 + (1-x)^3 = 1 \Rightarrow x^3 + 1 - 3x + 3x^2 - x^3 = 1 \Rightarrow x^2 - x = 0 \Rightarrow x = 0, 1 \Rightarrow (x, y) = (0, 1) (1, 0)$$

4.

$$n(M) = 23, n(P) = 24, n(C) = 19$$

$$n(M \cap P) = 12, n(M \cap C) = 9, n(P \cap C) = 7$$

$$n(M \cap P \cap C) = 4$$

$$n(M \cap P' \cap C') = n[M \cap (P \cup C)'] = n(M) - n(M \cap (P \cup C)) = n(M) - n[(M \cap P) \cup (M \cap C)]$$

$$= n(M) - n(M \cap P) - n(M \cap C) + n(M \cap P \cap C) = 23 - 12 - 9 + 4 = 27 - 21 = 6$$

$$n(P \cap M' \cap C) = n[P \cap (M \cup C)']$$

$$= n(P) - n[P \cap (M \cup C)] = n(P) - n[P \cap M \cup (P \cap C)] = n(P) - n(P \cap M) - n(P \cap C) + n(P \cap M \cap C)$$

$$= 24 - 12 - 7 + 4 = 9$$

$$n(C \cap M' \cap P') = n(C) - n(C \cap P) - n(C \cap M) + n(C \cap P \cap M)$$

$$= 19 - 7 - 9 + 4 = 23 - 16 = 7$$





$$5. \quad c(a-b) = a(b-c) \Rightarrow ac - bc = ab - ac \Rightarrow 2ac = ab + bc$$

$$\Rightarrow \frac{2ac}{b} = a + c \Rightarrow \frac{2ac}{a+c} = b$$

$$\text{Now } \frac{\log(a+c) + \log(a+c-2b)}{\log(a-c)} = \frac{\log(a+c) + \log\left(a+c - \frac{4ac}{a+c}\right)}{\log(a-c)}$$

$$= \frac{\log(a+c) + 2\log(a-c) - \log(a+c)}{\log(a-c)} = 2$$

$$6. \quad \log_b a \cdot \log_c a + \log_a b \cdot \log_c b + \log_a c \cdot \log_b c = 3$$

$$\Rightarrow \frac{(\log a)^2}{\log b \cdot \log c} + \frac{(\log b)^2}{\log a \cdot \log c} + \frac{(\log c)^2}{\log a \cdot \log b} = 3$$

$$\Rightarrow (\log a)^3 + (\log b)^3 + (\log c)^3 = 3 \log a \cdot \log b \cdot \log c$$

$$\Rightarrow \log a + \log b + \log c = 0 \quad (\because a, b, c \text{ are distinct})$$

$$\Rightarrow \log abc = 0 \Rightarrow abc = 1$$

$$7. \quad A = \log_{16} 4 = \frac{1}{2}$$

$$B = \log_3 9 = 2$$

$$\therefore 4^{1/2} + 9^2 = 10^{\log_x 83} \Rightarrow 83 = 83^{\log_x 10} \Rightarrow x = 10$$

$$8. \quad b^2 = a^3 = k \Rightarrow b = k^{1/2}, a = k^{1/3} \quad \text{and} \quad d^4 = c^5 = \lambda \Rightarrow c = \lambda^{1/5}, d = \lambda^{1/4}$$

$$\text{Now } a - c = 9 \Rightarrow k^{1/3} - \lambda^{1/5} = 9$$

$$\text{Let } k^{1/3} = 25, \lambda^{1/5} = 16 \Rightarrow k = 25^3 = 5^6 \text{ and } \lambda = 16^5 = 2^{20}$$

$$\Rightarrow b = (5^6)^{1/2} = 125, d = (2^{20})^{1/4} = 32$$

$$\text{Now } a - c = 9 \Rightarrow k^{1/3} - \lambda^{1/5} = 9$$

$$\text{let } k^{1/3} = 25, \lambda^{1/5} = 16$$

$$\Rightarrow b = (5^6)^{1/2} = 125, d = (2^{20})^{1/4} = 32$$

$$\text{Now } b - d = 125 - 32 = 93$$

$$\frac{b+d}{a+c} = \frac{157}{41} = 03.829$$

$$9. \quad \log_{10} (2x^2 - 21x + 50) = 2$$

$$(i) \quad 2x^2 - 21x + 50 = 100$$

$$\Rightarrow 2x^2 - 21x - 50 = 0 \Rightarrow x = -2, \frac{25}{2}$$

$$10. \quad \text{Domain } x-1 > 0 \text{ and } x+1 > 0 \text{ and } y-x > 0$$

$$x > 1 \quad x > -1 \quad x < 7$$

$$\Rightarrow x \in (1, 7) \quad \dots\dots\dots(i)$$

$$-\log_2 (x-1) - \log_2 (x+1) = 1 + \log_{2^{-1/2}} (7-x)$$

$$-\log_2 (x^2 - 1) + \log_2 (7-x)^2 = 1 \quad ; \quad \log_2 \frac{(7-x)^2}{x^2 - 1} = 1 \Rightarrow \frac{(7-x)^2}{x^2 - 1} = 2$$

$$\Rightarrow x^2 + 14x - 51 = 0 \quad ; \quad (x+17)(x-3) = 0$$

$$x = 3 \text{ or } x = -17 \text{ (rejected)} \quad ; \quad x = 3$$

$$11. \quad \log_{10}^2 x + \log_{10} x^2 = \log_{10}^2 2 - 1 \quad ; \quad \log_{10}^2 x + 2 \log_{10} x + 1 = \log_{10}^2 2$$

$$\Rightarrow (\log_{10} x + 1)^2 = \log_{10}^2 2$$

$$\log_{10} x + 1 = \pm \log_{10} 2 \quad ; \quad x = \frac{1}{20} \text{ and } \frac{1}{5}$$





12. $\frac{2009}{2010}x = (2009)^{\log_x(2010)}$

$\Rightarrow$ Taking log both sides w.r.t. x

$$\log_x 2009 + 1 - \log_x 2010 = \log_x 2010 \cdot \log_x 2009 \Rightarrow \frac{\ell n 2009}{\ell n x} + 1 - \frac{\ell n 2010}{\ell n x} = \frac{\ell n 2009 \ell n 2010}{(\ell n x)^2}$$

$$\Rightarrow \ell n \left(\frac{2009}{2010} \right) \cdot \ell n x + (\ell n x)^2 = \ell n 2009 \ell n 2010$$

Let $\ell n x = t$, $t^2 + \ell n \left(\frac{2009}{2010} \right) t - \ell n 2009 \ell n 2010 = 0$

sum of roots $= -\ell n \left(\frac{2009}{2010} \right) \Rightarrow \ell n x_1 + \ell n x_2 = -\ell n \left(\frac{2009}{2010} \right) = \ell n \left(\frac{2010}{2009} \right)$

$$\Rightarrow x_1 x_2 = \frac{2010}{2009}$$

$m = 2010, n = 2009 \Rightarrow m - n = 1$

13. $(\log x)^2 - \log x - 2 \geq 0$

$x > 0$ (i)

$(\log x - 2)(\log x + 1) \geq 0$

$\Rightarrow \log x \leq -1$ or $\log x \geq 2$

$\Rightarrow x \leq \frac{1}{10}$ or $x \geq 100$ (ii)

$(i) \cap (ii) \Rightarrow x \in \left(0, \frac{1}{10} \right] \cup [100, \infty)$

14. $\frac{x+1}{x+2} \Rightarrow x \in (-\infty, -2) \cup (-1, \infty)$

and $x + 3 > 0 \Rightarrow x > -3$

If $\log_4 \left(\frac{x+1}{x+2} \right) > \log_4 (x+3)$

$$\Rightarrow \frac{x+1}{x+2} > x+3 \Rightarrow \frac{x+1 - (x+3)(x+2)}{(x+1)} > 0$$

$$\Rightarrow \frac{x+1 - x^2 - 5x - 6}{x+1} > 0 \Rightarrow \frac{x^2 + 4x + 5}{x+1} < 0$$

$\Rightarrow x > -1$

ans. $(-1, \infty) \Rightarrow -a = -1 \Rightarrow a = -1$

If $\log_4 \left(\frac{x+1}{x+2} \right) < \log_4 (x+3)$

$\frac{x+1}{x+2} < x+3$ then solution is $(-3, -1)$

15. $\log_{1/2}(x+5)^2 > \log_{1/2}(3x-1)^2$

$(x+5)^2 > 0 \Rightarrow x \in \mathbb{R} - \{-5\}$ (i)

$(3x-1)^2 > 0 \Rightarrow x \in \mathbb{R} - \left\{ \frac{1}{3} \right\}$ (ii)

$(x+5)^2 < (3x-1)^2$

$\Rightarrow 8x^2 - 16x - 24 > 0$

$\Rightarrow x^2 - 2x - 3 > 0$

$\Rightarrow (x-3)(x+1) > 0$

$\Rightarrow x \in (-\infty, -1) \cup (3, \infty)$ (iii)

$(i) \cap (ii) \cap (iii)$ gives

$(-\infty, -5) \cup (-5, -1) \cup (3, \infty)$

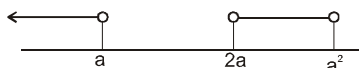
$p = -5, q = -5, r = -1, s = 3$





PART - III

1.



$$\text{so } a - 1 + a^2 - 2a - 1 = 18$$

$$a = 5, -4 \quad \therefore \quad a = 5$$

2.

$$N = \frac{\log_3 135}{\log_{15} 3} - \frac{\log_3 5}{\log_{405} 3} = (\log_3 27 + \log_3 5) - (\log_3 15) \log_3 5 \cdot \log_3 405$$

$$= (3 + \log_3 5) (1 + \log_3 5) - \log_3 5 \log_3 (81 \times 5) = (3 + \log_3 5) (1 + \log_3 5) - \log_3 5 (4 + \log_3 5) = 3$$

3.

$$(\log_5 x)^2 + \log_{5x} \frac{5}{x} = 1$$

$$\Rightarrow (\log_5 x)^2 + \log_{5x} 5 - \log_{5x} x = 1$$

$$\Rightarrow (\log_5 x)^2 + \frac{\log_5 5}{\log_5 5 + \log_5 x} - \frac{\log_5 x}{\log_5 5 + \log_5 x} = 1$$

$$\Rightarrow (\log_5 x)^2 + \frac{1}{1 + \log_5 x} - \frac{\log_5 x}{1 + \log_5 x} = 1$$

$$\text{Let } \log_5 x = t$$

$$\therefore t^2 + \frac{1}{1+t} - \frac{t}{1+t} = 1 \Rightarrow \frac{t^2(1+t) + 1 - t}{1+t} = 1 \Rightarrow t^3 + t^2 + 1 - t = 1 + t$$

$$t^3 + t^2 - 2t = 0 \quad ; \quad t(t^2 + t - 2) = 0 \quad ; \quad t(t-1)(t+2) = 0$$

$$t = 0, 1, -2$$

$$\therefore \log_5 x = 0, 1, -2$$

$$\therefore x = 1, 5, \frac{1}{25}$$

4.

$$\log_{x^2} 16 + \log_{2x} 64 = 3 \Rightarrow 4 \log_{x^2} 2 + 6 \log_{2x} 2 = 3$$

$$\Rightarrow \frac{4}{\log_2 x^2} + \frac{6}{\log_2 2x} = 3 \Rightarrow \frac{2}{\log_2 x} + \frac{6}{1 + \log_2 x} = 3 \text{ but } \log_2 x = t$$

$$\therefore \frac{2}{t} + \frac{6}{1+t} = 3 \Rightarrow 2 + 2t + 6t = 3t + 3t^2$$

$$\Rightarrow 3t^2 - 5t - 2 = 0 \Rightarrow 3t^2 - 5t - 2 = 0 \Rightarrow (3t+1)(t-2) = 0$$

$$\Rightarrow t = -\frac{1}{3}, t = 2 \Rightarrow \log_2 x = -\frac{1}{3} \quad \log_2 x = 2$$

$$\Rightarrow x = 2^{-1/3} \quad x = 4 = \frac{1}{2^{1/3}}$$

5.

$$x^{[(\log_3 x)^2 - \frac{9}{2} \log_3 x + 5]} = 3 \sqrt{3} \Rightarrow (\log_3 x)^3 - \frac{9}{2} \log_3 x + 5 = \log_x 3 \sqrt{3}$$

$$\Rightarrow (\log_3 x)^2 - \frac{9}{2} \log_3 x + 5 = \frac{3}{2} \log_x 3$$

$$\text{Let } \log_3 x = t \Rightarrow t^2 - \frac{9}{2} t + 5 = \frac{3}{2t} \Rightarrow 2t^3 - 9t^2 + 10t - 3 = 0$$

$$t = 1 \text{ satisfies it } 2t^3 - 9t^2 + 10t - 3 = 2t^2(t-1) - 7t(t-1) + 3(t-1)$$

$$= (t-1)(2t^2 - 7t + 3) = (t-1)(2t-1)(t-3) \Rightarrow t = 1 \quad t = \frac{1}{2} \quad t = 3$$

$$\Rightarrow \log_3 x = 1 \quad \log_3 x = \frac{1}{2} \quad \log_3 x = 3$$

$$\Rightarrow x = 3 \quad x = 3^{1/2} \quad x = 27.$$





6. $\log_3 x + \log_3 y = 2 + \log_3 2$, $\log_{27}(x + y) = \frac{2}{3}$

$$\Rightarrow \log_3 xy = \log_3 9 + \log_3 2,$$

$$\Rightarrow x + y = (27)^{2/3}$$

$$\Rightarrow xy = 18,$$

$$\Rightarrow x + y = 9$$

$$\Rightarrow x = 6 \text{ or } x = 3, y = 3, y = 6 \quad \text{as } x > 0, y > 0.$$

7. $(\log_{10} 8)x^2 - (\log_{10} 5)x + x - 2\log_{10} 2 = 0$

(A) sum of roots = $\frac{-(1 - \log_{10} 5)}{\log_{10} 8} = \frac{-\log_{10} \left(\frac{10}{5}\right)}{\log_{10} 8} = \frac{-\log_{10} 2}{3\log_{10} 2} = \frac{-1}{3}$ rational

(B) Product of roots = $\frac{-2\log_{10} 2}{\log_{10} 8} = \frac{-2}{3}$

(C) sum of coefficient = $\log_{10} 8 - \log_{10} 5 + 1 - \log_{10} 4 = \log_{10} \left(\frac{8 \times 10}{5 \times 4}\right)$
 $= \log_{10} 4 = \text{irrational}$

(D) Discriminant = $(\log_{10} 2)^2 - 4\log_{10} 8(-2\log_{10} 2) = (\log_{10} 2)^2 + 24(\log_{10} 2)^2$
 $= 25(\log_{10} 2)^2 = (5\log_{10} 2)^2$ irrational.

8. $x = a^b$

(D) If a is rational & b is rational then x may be rational
 e.g. $= 2^2$

(C) $(\sqrt{2})^4$

(B) $(2)^{\log_2 3}$

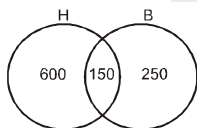
(A) $(\sqrt{2})^{\log_2 9}$

9. $\log_2 3 > 1, \log_{12} 10 < 1 \Rightarrow \log_2 3 > \log_{12} 10$
 $\log_5 5 < 1, \log_7 8 > 1 \Rightarrow \log_5 5 < \log_7 8$
 $\log_3 26 < 3, \log_2 9 > 3 \Rightarrow \log_3 26 < \log_2 9$
 $\log_{16} 15 < 1, \log_{10} 11 > 1 \Rightarrow \log_{16} 15 < \log_{10} 11$

10. $\frac{1}{2} \leq \log_{1/10} x \leq 2 \Rightarrow \frac{1}{100} \leq x \leq \frac{1}{\sqrt{10}}$

PART - IV

Sol. (Q 1 to 3)



$$n(H \cup B) = n(H) + n(B) - n(H \cap B)$$

$$1000 = 750 + 400 - n(H \cap B) = 150$$

Now $n(\text{only hindi}) = n(H) - n(H \cap B) = 750 - 150 = 600$

$n(\text{only bengali}) = n(B) - n(H \cap B)$

$400 - 150 = 250$

Sol. (Q 4 to 6) Let $\log_4 x = t \Rightarrow 1 + t + 4(5 - 4t) = 3(5 - 4t)(1 + t)$

$$21 - 15t = 15 + 3t - 12t^2$$

$$\Rightarrow 12t^2 - 18t + 6 = 0 \Rightarrow 2t^2 - 3t + 1 = 0$$

$$t = 1, t = 1/2 \Rightarrow x = 4' \text{ or } x = 4^{1/2} = 2$$





$$A = \text{sum of roots} = 4 + 2 = 6$$

$$\begin{aligned} \text{for B: } 2^m = 3, 3^n = 4 &\Rightarrow m = \log_2 3, \Rightarrow n = \log_3 4 \\ &\Rightarrow \text{Now } mn = \log_2 3 \cdot (\log_3 2^2) = 2 \quad B = 2 \end{aligned}$$

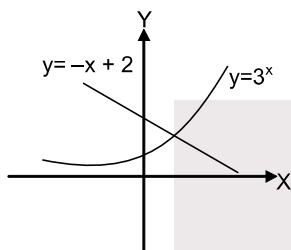
$$\begin{aligned} \text{for C: } \frac{1 - \log_3 x}{1 + \log_3 x} + (\log_3 x)^2 &= 1 \Rightarrow 1 - t + t^2(1 + t) = 1 + t \\ &\Rightarrow t^2 + t^3 - t = t \\ t^3 + t^2 - 2t &= 0 \Rightarrow t(t^2 + t - 2) = 0 \Rightarrow t = 0, t = -2, t = 1 \\ x = 3^t &\Rightarrow x = 3^0 = 1, x = 3^{-2} = 1/9 \end{aligned}$$

$$\text{sum of integral root} = C = 1 + 3 = 4$$

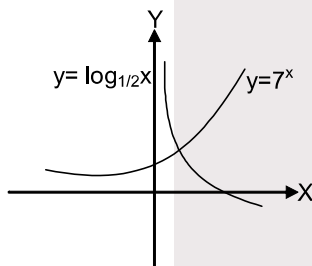
$$(4) \quad A + B = 6 + 2 = 8 \quad (5) \quad B + C = 2 + 4 = 6$$

$$(6) \quad A + C \div B = 6 + 4 \div 2 = 6 + 2 = 8$$

8.



9.



EXERCISE # 3

PART - I

1. Obvious

2. Obvious

$$\begin{aligned} 3. \quad 2 \log_2 \log_2 x + \log_{1/2} \log_2 (2\sqrt{2}x) &= 1 \Rightarrow \log_2 (\log_2 x)^2 - \log_2 \log_2 (2\sqrt{2}x) = 1 \\ \Rightarrow \log_2 \frac{(\log_2 x)^2}{\log_2 (2\sqrt{2}x)} &= 1 \Rightarrow \frac{(\log_2 x)^2}{\frac{3}{2} + \log_2 x} = 2 \end{aligned}$$

$$\text{let } \log_2 x = y$$

$$\therefore y^2 - 2y - 3 = 0$$

$$\therefore y = 3, -1$$

$$\text{but } \log_2 x > 0$$

$$\therefore \log_2 x = -1 \text{ is not possible} \Rightarrow x = 8$$

$$4. \quad \log_{3/4} \log_8 (x^2 + 7) + \log_{1/2} \log_{1/4} (x^2 + 7)^{-1} = -2$$

$$\Rightarrow \log_{3/4} \frac{1}{3} \log_2 (x^2 + 7) - \log_2 \frac{\log_2 (x^2 + 7)}{2} = -2$$

$$\text{let } \log_2 (x^2 + 7) = t$$





$$\Rightarrow \log_{3/4} \frac{t}{3} - \log_2 \frac{t}{2} + 2 = 0 \Rightarrow \log_{3/4} \frac{t}{3} + 1 - \left(\log_2 \frac{t}{2} - 1 \right) = 0$$

$$\Rightarrow \log_{3/4} \frac{t}{4} = \log_2 \frac{t}{4} \Rightarrow \frac{t}{4} = 1 \Rightarrow t = 4$$

$$\therefore \log_2 (x^2 + 7) = 4$$

this gives $x = \pm 3$

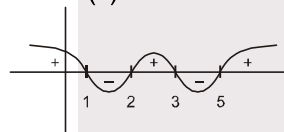
$$\begin{aligned} 5. \quad \frac{1}{2} \log_2 (x-1) &= \log_2 (x-3) \Rightarrow \sqrt{x-1} = x-3 \\ (x-1) &= x^2 - 6x + 9 \Rightarrow x^2 - 7x + 10 = 0 \\ (x-5)(x-2) &= 0 \quad \text{but } x \neq 2 \\ \therefore x &= 5 \end{aligned}$$

$$6. \quad \text{Given } f(x) = \frac{x^2 - 6x + 5}{x^2 - 5x + 6}$$

$$(i) \quad \text{when } 0 < f(x) < 1 \quad \text{then } 0 < \frac{x^2 - 6x + 5}{x^2 - 5x + 6} < 1$$

$$\begin{aligned} \text{So } \frac{(x-5)(x-1)}{(x-2)(x-3)} > 0 \text{ and } \frac{x^2 - 6x + 5}{x^2 - 5x + 6} - 1 < 0 &\Rightarrow \frac{(x+1)}{(x-2)(x-3)} > 0 \\ \Rightarrow x \in (-1, 1) \cup (5, \infty) \end{aligned}$$

$$(ii) \quad \text{when } f(x) < 0$$

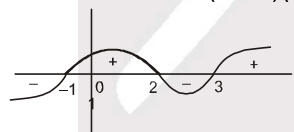


$$\frac{(x-1)(x-5)}{(x-2)(x-3)} < 0$$

$$\Rightarrow x \in (1, 2) \cup (3, 5)$$

$$(iii) \quad f(x) > 0, x \in (-\infty, 1) \cup (2, 3) \cup (5, \infty)$$

$$(iv) \quad f(x) < 1 \Rightarrow \frac{x+1}{(x-2)(x-3)} > 0$$



$$x \in (-1, 2) \cup (3, \infty)$$

- (A) $-1 < x < 1$, $f(x)$ satisfies p, q, s
 (B) $1 < x < 2$, $f(x)$ satisfies q, s
 (C) $3 < x < 5$, $f(x)$ satisfies q, s
 (D) $x > 5$, $f(x)$ satisfies p, r, s

$$7. \quad (2x)^{\ln 2} = (3y)^{\ln 3}$$

$$\Rightarrow \ln 2 \ln(2x) = \ln 3 \ln(3y) = \ln 3 (\ln 3 + \ln y) \quad \dots\dots\dots (1)$$

$$\text{also } 3^{\ln x} = 2^{\ln y} \Rightarrow \ln x \ln 3 = \ln y \ln 2 \quad \dots\dots\dots (2)$$

$$\text{by (1)} \Rightarrow \ln 2 \ln(2x) = \ln 3 (\ln 3 + \ln y) \Rightarrow \ln 2 \cdot \ln(2x) = \ln 3 \left\{ \ln 3 + \frac{\ln x \ln 3}{\ln 2} \right\}$$

$$\Rightarrow \ln^2 2 \ln 2x = \ln^2 3 (\ln 2 + \ln x) \Rightarrow (\ln^2 2 - \ln^2 3) (\ln 2x) = 0 \Rightarrow \ln 2x = 0 \Rightarrow x = \frac{1}{2}$$

$$8. \quad \text{Let } \sqrt{4 - \frac{1}{3\sqrt{2}}} \sqrt{4 - \frac{1}{3\sqrt{2}}} \dots\dots\dots = t \Rightarrow \sqrt{4 - \frac{1}{3\sqrt{2}}} t = t \Rightarrow 4 - \frac{1}{3\sqrt{2}} t = t^2 \Rightarrow$$





$$t^2 + \frac{1}{3\sqrt{2}} t - 4 = 0 \Rightarrow 3\sqrt{2} t^2 + t - 12\sqrt{2} = 0 \Rightarrow t = \frac{-1 \pm \sqrt{1 + 4 \times 3\sqrt{2} \times 12\sqrt{2}}}{2 \times 3\sqrt{2}} = \frac{-1 \pm 17}{2 \times 3\sqrt{2}}$$

$$t = \frac{16}{6\sqrt{2}}, \frac{-18}{6\sqrt{2}} \Rightarrow t = \frac{8}{3\sqrt{2}}, \frac{-3}{\sqrt{2}} \text{ and } \frac{-3}{\sqrt{2}} \text{ is rejected}$$

$$\text{so } 6 + \log_{3/2} \left(\frac{1}{3\sqrt{2}} \times \frac{8}{3\sqrt{2}} \right) = 6 + \log_{3/2} \left(\frac{4}{9} \right) = 6 + \log_{3/2} \left(\left(\frac{2}{3} \right)^2 \right) = 6 - 2 = 4$$

$$9^*. \quad 3^x = 4^{x-1} \Rightarrow x = (x-1) \log_3 4 \Rightarrow x(1 - 2\log_3 2) = -2\log_3 2$$

$$x = \frac{2\log_3 2}{2\log_3 2 - 1} \quad \text{Ans. (A)}$$

$$\text{Again } x \log_2 3 = (x-1) \cdot 2 \Rightarrow x(\log_2 3 - 2) = -2 \Rightarrow x = \frac{2}{2 - \log_2 3} \quad \text{Ans. (B)}$$

$$x = \frac{1}{1 - \frac{1}{2} \log_2 3} = \frac{1}{1 - \log_4 3} \quad \text{Ans. (C)}$$

$$10. \quad \left((\log_2 9)^2 \right)^{\frac{1}{\log_2 (\log_2 9)}} \times (\sqrt{7})^{\log_7 4} \\ (\log_2 9)^{2 \log_2 (\log_2 9)^2} \cdot (2) = 4.2 = 8$$

PART - II

1. Obvious	2. Obvious	3. Obvious	4. Obvious
5. Obvious	6. Obvious	7. Obvious	8. Obvious

$$9. \quad \begin{aligned} &\text{We have, } A \cup B = A \cup C \Rightarrow (A \cap C) \cup (B \cap C) = C \\ &\Rightarrow (A \cap B) \cup (B \cap C) = C \quad \dots(i) \\ &\text{Again, } A \cup B = A \cup C \Rightarrow (A \cup B) \cap B = (A \cup C) \cap B \Rightarrow B = (A \cap B) \cup (C \cap B) \\ &\Rightarrow (A \cap B) \cup (C \cap B) = B \Rightarrow (A \cap B) \cup (B \cap C) = B \quad \dots(ii) \\ &\text{From (i) and (ii), we get } B = C \end{aligned}$$

10. Every element has 3 options. Either set Y or set Z or none
so number of ordered pairs = 3^5

$$11. \quad \begin{aligned} X &= \{0, 9, \dots, 4^n - 3n - 1\} \\ Y &= \{0, 9, \dots, 9(n-1)\} \\ \text{Now } 4^n - 3n - 1 &= (3+1)^n - 3n - 1 \\ &= 3^n + n \cdot 3^{n-1} + \dots + {}^nC_2 \cdot 9. \end{aligned}$$

is a multiple of 9.

Also Y consists of all multiples of '9' from 0, 9,

Hence all values of X are subset of values of Y.

Thus $X \cup Y = Y$

$$12. \quad (x^2 - 5x + 5)^{x^2 + 4x - 60} = 1$$

$$x^2 - 5x + 5 = 1$$

$$x^2 - 5x + 4 = 0$$

$$x = 1, x = 4$$

$$\text{at } x=2, x^2 + 4x - 60 = -48$$

$$\therefore x=2 \text{ is valid}$$

$$\text{at } x=3, x^2 + 4x - 60 = -39 \text{ (odd)}$$

$$\therefore x=3 \text{ is invalid}$$

$$x = 1, 2, 4, 6, -10$$

$$x^2 + 4x - 60 = 0$$

$$x = -10, x = 6$$

$$x^2 - 5x + 5 = -1$$

$$x^2 - 5x + 6 = 0$$

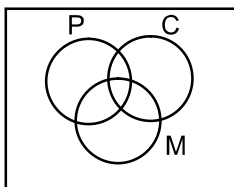
$$x = 2, 3$$





$$13. \quad n(P) = \left[\frac{140}{3} \right] = 46$$

$$n(C) = \left[\frac{140}{5} \right] = 28$$



$$n(M) = \left[\frac{140}{2} \right] = 70$$

$$n(P \cup C \cup M) = n(P) + n(C) + n(M) - n(P \cap C) - n(C \cap M) - n(M \cap P) + n(P \cap M \cap C)$$

$$= 46 + 28 + 70 - \left[\frac{140}{15} \right] - \left[\frac{140}{10} \right] - \left[\frac{140}{6} \right] + \left[\frac{140}{30} \right]$$

$$= 144 - 9 - 14 - 23 + 4 = 102$$

$$\text{so required number of student} = 140 - 102 = 38$$

$$14. \quad \begin{aligned} n(A \cup B) &= n(A) + n(B) - n(A \cap B) \\ &= 25 + 7 - 3 \\ &= 29 \end{aligned}$$

